Journal of Workplace Learning

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A safety ethogram for military exercises: a practitioner framework for governance, drift, intervention, and learning

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Received 30 March 2026
Revised 21 June 2026
Accepted 22 June 2026

Abstract

Purpose – This paper aims to present a practitioner-oriented Safety Ethogram for military exercises. Its purpose is to make safety-related learning practices easier to observe, discuss and improve during planning, execution, debrief and after-action review.

Design/methodology/approach – This paper draws on qualitative content analysis of interview material on safety planning, coordination, deviation, intervention and learning in military exercises. Of 42 interview transcript files, 38 were readable and analysed through 74 traceable meaning units using a structured coding framework focused on interview-described safety practices.

Findings – Four recurring areas of practice were identified: formal safety governance and risk work, deviation and drift from intended practice, intervention to correct or redirect exercise activity and learning through experience and institutional memory. Together, these areas show that safety in exercises is not maintained by rules alone. It also depends on how governance is translated into action, how drift is recognised, whether intervention is legitimate and usable and whether lessons identified are carried forward into future planning.

Research limitations/implications – The framework is based on interview-described practice rather than direct observation. It should therefore be read as a practice-informed tool that requires further testing in live exercise settings.

Practical implications – The safety ethogram offers a practical vocabulary for those who plan, lead, observe and evaluate exercises. It can be used to strengthen observation, structure debriefs, diagnose drift and improve the link between after-action learning and future safety governance.

Originality/value – This paper contributes a practice-based framework for examining how safety is enacted in military exercises as workplaces. Rather than treating safety as an abstract cultural property, it makes visible the concrete patterns through which safety is supported, weakened, corrected and learned.

Keywords Military exercises, Safety, Workplace learning, After-action review, Organisational learning, Qualitative content analysis

Paper type Practitioner paper

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Journal of Workplace Learning
Vol. 38 No. 9, 2026
pp. 207-224
Emerald Publishing Limited
1366-5626
DOI 10.1108/JWL-03-2026-0167

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Introduction

Military exercises as safety-critical workplace-learning environments

Military exercises are often framed as preparation for future operations, but they also function as temporary, high-tempo workplaces in which personnel coordinate activity, make decisions under pressure and carry out tasks with real consequences for safety, learning and professional credibility. In workplace-learning terms, exercises are deliberate learning interventions: they generate experience that can be examined against doctrine, standards and professional judgement, then translated into improved capability. Yet exercises must balance realism and risk control, becoming sufficiently demanding to stretch competence without normalising unsafe practice. This tension is especially visible in two-sided exercises, where competition, performance evaluation and mission primacy can increase tempo and shape safety-relevant behaviour (Schüler, 2022a).

Workplace-learning research suggests that learning depends on the interplay between workplace affordances and individual or collective engagement (Billett, 2001; Billett, 2004). In military exercises, what participants learn about safety is shaped not only by rules, plans and formal objectives, but also by situated practices repeatedly enacted and reinforced as workable under tempo, interdependence, hierarchy and evaluation pressure.

Safety science, drift and the gap between plans and practice

Safety-science research has long shown that safety-critical work is not maintained by rules alone. Formal plans, risk assessments and procedures describe work-as-imagined, while activity in complex settings is shaped by constraints, competing goals, local adaptation and work-as-done (Hollnagel, 2006; Hollnagel, 2014). Deviation is therefore not necessarily a simple failure of compliance. Some departures from plan may represent competent adaptation, whereas others may indicate unsafe drift, normalised shortcuts or erosion of safety margins (Dekker, 2011; Rasmussen, 1997; Vaughan, 1996). This distinction is central to military exercises because exercises are deliberately designed to create pressure, uncertainty and friction while still remaining bounded by peacetime safety requirements.

High reliability Organisation theory and resilience engineering are relevant because they direct attention to anticipation, monitoring, response, learning and the organisational conditions that allow weak signals and emerging risks to be noticed before they produce harm (Weick and Sutcliffe, 2007; Hollnagel, 2010). Concepts such as mindful organising, adaptive capacity and work-as-imagined/work-as-done help explain why formal safety governance may be necessary but insufficient. They also highlight the importance of intervention legitimacy: the practical and social possibility of stopping, resetting, redirecting or questioning an exercise trajectory when it becomes unsafe or educationally misleading.

However, these safety-science perspectives do not fully resolve the workplace-learning problem addressed here. Even when drift, adaptation or intervention is recognised, it remains necessary to understand how such practices become visible, discussable, legitimate and institutionally consequential. In military exercises, the issue is not only whether safety-relevant events occur, but whether they are noticed, examined and translated into future planning, debriefing and after-action review.

Workplace learning, reflection and institutional memory

From a learning perspective, experience does not automatically become learning. Dewey's account of reflective thinking treats experience as a starting point that must be subjected to inquiry if it is to become educative rather than merely repetitive (Dewey, 1998). Schön's work on reflective practice similarly highlights how professional knowledge develops through engagement with uncertainty, surprise and breakdowns in routine (Schön, 1983). In

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organisational settings, the issue is therefore not merely whether events occurred, but whether they were interpreted, discussed and incorporated into future action.

Organisational learning research further suggests that safety improvement depends on more than collecting lessons. Lessons identified must become lessons implemented. Reporting systems, after-action reviews, investigations and safety committees may support learning, but they may also become formal routines that document experience without substantially changing future practice (Carroll *et al.*, 2002; Cannon and Edmondson, 2005; Argyris, 1976). Institutional memory therefore has a dual character: it can preserve hard-won safety knowledge, but it can also reproduce outdated assumptions, "old truths" or exercise-specific habits that are no longer aligned with current risk conditions. Recent work on organisational unlearning similarly highlights that established routines may need to be questioned or abandoned before new learning can take effect (Klammer and Gueldenberg, 2019; Maccioni *et al.*, 2024).

The gap addressed by this study

Despite extensive work on workplace learning, safety science and organisational learning, a practical and conceptual gap remains. Safety in military exercises is often discussed at the level of rules, risk assessments, culture or outcomes, while the practice-level patterns through which safety is described, negotiated, challenged, corrected and remembered are less visible. Put differently, it is one thing to argue that exercises should promote safe learning; it is another to specify what kinds of safety-relevant practices should be noticed in planning, execution, debrief and after-action review.

This gap matters because claims about safety learning can become abstract if they are not anchored in concrete practices. It also matters practically because debriefs and after-action reviews may become dominated by outcomes, anecdotes, compliance language or official success narratives rather than by systematic examination of how risk was handled during exercise activity. Prior work on military exercises has shown that competitive dynamics may alter learning conditions and safety-relevant behaviour, while earlier ethogram-based work has suggested that exercise behaviours can be classified through a functional inventory of what participants and controllers do in exercises (Schüler and Bjurström, 2023; Schüler, 2022a). The present study builds on that logic but narrows its focus to safety-relevant practice.

This study therefore develops an interview-derived Safety Ethogram for military exercises. In this paper, an ethogram is understood as a structured catalogue of safety-relevant practices, routines and interaction patterns that can guide future observation, structured debrief and after-action review. Because the study is based on retrospective interview accounts, the framework should not be read as a direct observational record of exercise behaviour or as a behavioural frequency measure. Rather, it classifies how military personnel describe safety-relevant practices across exercise work and translates these descriptions into a practitioner-oriented heuristic.

Contribution and aim

The contribution of the paper is twofold. First, it contributes to workplace-learning research by specifying an interview-derived framework through which safety learning in military exercises can be examined as a coupled process involving formal governance, deviation and drift, intervention and institutional memory. Secondly, it contributes to safety-science and military exercise practice by translating concerns about work-as-imagined/work-as-done, adaptive deviation, drift and intervention legitimacy into a debrief-oriented framework that can support more systematic reflection and after-action learning.

The aim of the study is therefore to develop an interview-derived Safety Ethogram for military exercises, conceptualised as safety-critical workplace-learning environments.

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Specifically, the study examines how personnel describe safety-relevant practices related to formal safety governance and risk work, deviation and drift, intervention shaping exercise trajectory and learning, experience use and institutional memory. The resulting framework is intended to support more disciplined observation, discussion and future validation of safety-relevant practice in military exercise settings.

Method

Design

This study used qualitative content analysis to develop an interview-derived Safety Ethogram for military exercises. The analysis followed a directed content analysis strategy, in which an initial coding frame derived from prior ethogram-based work was applied to interview material and then refined through operational definitions, inclusion criteria, exclusion criteria and boundary rules (Hsieh and Shannon, 2005; Elo and Kyngas, 2008). The reporting approach was structured with reference to the Standards for Reporting Qualitative Research (SRQR) (O'Brien *et al.*, 2014).

The study did not seek to generate an entirely inductive typology from first principles. Rather, it aimed to re-specify an existing ethogram logic for a more focused analytical purpose: the examination of safety-relevant practices in military exercise work. The design was therefore appropriate for developing a practice-oriented classification framework while retaining evidentiary discipline between the interview material, the meaning units and the final reported behaviour classes.

Because the study is based on retrospective interview accounts, the reported classes should be understood as interview-described safety practices rather than as direct observations of exercise behaviour. The Safety Ethogram is therefore not presented as a behavioural frequency measure or as an empirically observed sequence of action. It is used as a heuristic framework for classifying how personnel describe safety-relevant practices, routines and interaction patterns in relation to planning, execution, intervention and learning.

Setting and context

The study was conducted in a Swedish military context in which exercises are used both to develop operational capability and to generate learning through structured preparation, execution, debrief and after-action review. Military exercises are safety-critical workplace-learning environments because they combine formal planning, hierarchical authority, mission-oriented performance demands, realistic training conditions and peacetime safety obligations. This context is analytically important because safety-related accounts may be shaped by rank relations, professional norms, organisational accountability and the perceived legitimacy of discussing deviation, rule-bending, intervention or learning failure.

To reduce the risk of identifying individuals, units, locations or operational arrangements, the manuscript reports the setting at an aggregated level. Specific unit identifiers, exercise names, locations and operationally sensitive details were removed or generalised during transcription, extraction and reporting.

Participants and sample characteristics

The corpus comprised interview material from personnel with experience relevant to military exercises, safety planning, execution, intervention or learning. Participants were recruited to capture accounts from individuals positioned differently in relation to exercise work, including planning, safety governance, execution, exercise control and organisational learning.

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Table 1 summarises the readable interview files at an aggregated level. Of the 42 original interview files, four were corrupted or unreadable, leaving 38 files available for analysis. The readable sample was dominated by experienced personnel, with 32 participants having more than 15 years of experience and six having less than 10 years. The broad organisational distribution was weighted towards Army/land-force contexts, but also included maritime, aviation, logistics, technical, headquarters and cross-functional roles. This composition is relevant for transferability: the material provides strong access to command, staff, planning, governance and safety-related perspectives, while field-level execution perspectives are less prominent.

Materials and data corpus

The corpus comprised 42 interview transcript files, labelled X01-X42. Four files were corrupted and unreadable, leaving 38 files available for analysis. The analysis was based on a consolidated extraction table containing 74 traceable meaning units, each linked to a file identifier, local context label, extract or de-identified paraphrase and one or more ethogram codes. Meaning units were selected when they described safety-relevant practices, routines, interaction patterns or organisational processes associated with military exercise work (see Table 2). The corpus should therefore be understood as a source of interview-described safety practice, not as a direct observational record of exercise behaviour. To improve transparency, the full de-identified extraction table will be provided as supplementary material, subject to confidentiality, OPSEC and deductive disclosure checks.

Recruitment, consent and data collection

Recruitment relied on informal, researcher-initiated conversations during site visits rather than nomination through the chain of command. This approach was used to reduce perceived pressure to participate in a hierarchical organisational setting. Individuals who expressed interest were invited to participate on an opt-in basis and were informed that participation

Table 1. Aggregated participant characteristics

Sample characteristic	Aggregated category	*n*
Interview-file status	Original interview files	42
Interview-file status	Corrupted/unreadable files	4
Interview-file status	Readable files included in analysis	38
Experience band	Less than 10 years	6
Experience band	More than 15 years	32
Broad organisational context	Army/land forces	25
Broad organisational context	Navy/maritime	4
Broad organisational context	Air/aviation	3
Broad organisational context	Logistics, technical and support functions	4
Broad organisational context	Joint, headquarters and cross-functional roles	2
Broad role orientation	Command, staff and planning roles	18
Broad role orientation	Safety, medical, occupational safety and risk-related roles	7
Broad role orientation	Technical, human resources, information and support roles	6
Broad role orientation	Field-level execution, policing, aviation and instruction roles	7

Note(s): The table reports aggregated characteristics for the 38 readable interview files. Four files were corrupted or unreadable and excluded from meaning-unit extraction. Exact rank, unit, appointment and narrow specialist function are withheld to reduce deductive disclosure risk

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Table 2. Examples from meaning unit mapping

MU	File	Local context	Meaning unit (extract)	Theme
MU07	X11	Clarifying planning methods and responsibility	"…either you have the responsibility or you don't… explain… those who have responsibility… it's important to produce supporting documentation…"	E5
MU10	X13	Using formal routines and feedback loops	"…workplace-meeting minutes (APT)… common template… checkboxes… goes directly to the safety committee… feedback… learning…"	E5
MU11	X14	Describing shortcuts that persist	"…you cut corners… 'shortcuts'… quick fix… then you come home… 'you were never allowed to do that'… 'still a problem'…"	E6, E1
MU12	X14	Describing planning differing by level	"…at staff level… planning… good… out in the field… 'go fast and wrong'… at lower levels you're not given that time…"	E5, E1
MU21	X36	Describing goals forcing "no reset" and official success	"…the objectives have to be achieved… rolling back… 'never happens'… 'officially we always have goal fulfilment'…"	E4, E1
MU26	X40	Breaking rules already during planning	"…the power-line corridor… 'then you're breaking it already in the planning'…"	E5, E1
MU53	X29	Tracing the historical development of SMS from accidents	The safety process is described as historically grown (since the 1960s) due to many accidents; includes reporting + trend analysis. "…captured in the 60 s… many accidents… created… SMS… deviation system… trend analyses…"	E6
MU57	X30	Treating adaptive deviation ("break the plan") as competence	Normatively encourages breaking the plan when required; "must dare" to deviate. "…you must dare to break… put away the plan…"	E1
MU58	X30	Avoiding "plan worship" by balancing priorities	Planning is necessary but not to be "plan-worship"; balancing priorities. "…planning is important… but… not worship the plan…"	E5
MU72	X36	Using the "red card" as a stop-authority intervention tool	"Red card" capability to stop unsafe/incorrect exercise conduct (controller-like intervention).	E4
MU70	X37	Refusing to cut safety while training validity suffers (negative case)	They refuse to "cut safety," but admit training becomes wrong if compensating with full lighting due to missing equipment. "…we won't compromise safety… then full lighting… then you train wrong…"	E5

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was voluntary, that they could decline without explanation and that participation status would not be communicated to supervisors or senior military leadership.

Participants chose the interview time and location they regarded as most suitable and safest. Most interviews were conducted during duty time, based on participant preference and practical feasibility. Locations included a dedicated office on the installation intended to reduce routine oversight and civilian settings outside the installation. No compensation was provided.

The interviews were conducted in Swedish using a semi-structured format. Where possible, interviews were audio-recorded and transcribed *verbatim* by the researcher. The interview guide explored experiences of safety planning and coordination in training and exercises, including responsibility allocation, communication pathways, uncertainty, intervention and learning or follow-up after incidents and near misses. To elicit accounts of everyday safety practice rather than narrow assessments of formal compliance, interviews opened with a broad prompt inviting participants to reflect on what terms associated with "safety" meant in their work.

Analytical framework: adapting the ethogram logic

The analytical framework built on the peer-reviewed ICCRTS proceedings paper Navigating in the zoo (Schüler and Bjurström, 2023), which proposed an ethogram logic for classifying exercise behaviours through a functional inventory relevant to exercise conduct and outcome shaping. In the present study, that logic was re-specified for safety-related practice in military exercise work.

The original E-numbering was retained to preserve traceability to the source framework. To improve readability for readers unfamiliar with the earlier work, each code is referred to by both its label and functional meaning. The term "behaviour class" is used broadly. It includes not only discrete individual actions, but also safety-relevant routines, practices, interaction patterns and organisational processes through which exercise work is planned, adapted, interrupted or remembered.

The final interview-supported Safety Ethogram comprised four classes: formal safety governance and risk work (E5), deviation, drift and rule-bending (E1), intervention shaping exercise trajectory (E4) and learning, experience use and institutional memory (E6). Two candidate classes from the source framework, unauthorised equipment use and unauthorised information use, were not reported in the extracted meaning units and were therefore not included as empirically supported classes. This should not be interpreted as evidence that such behaviours are absent from military exercises, since practices carrying reputational, disciplinary or self-incriminating risk may be vulnerable to non-disclosure in hierarchical interview settings.

Coding procedure

Coding followed a structured, audit-trailed workflow consistent with directed qualitative content analysis (Hsieh and Shannon, 2005). First, the transcripts were read for familiarisation to identify passages describing safety as practice across planning, execution, intervention and learning. Secondly, relevant passages were extracted and segmented into meaning units. Each meaning unit was entered into the consolidated extraction table with a file identifier, local context label, extract or paraphrase and preliminary code.

Thirdly, meaning units were coded deductively to one or more ethogram classes using operational definitions derived from the source framework and refined for the safety emphasis of the present study. Multiple coding was allowed where a meaning unit linked more than one safety-relevant practice. For example, an account of shortcuts in planning

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could be coded as both formal safety governance and deviation or drift if it described a gap between documented requirements and enacted practice.

Fourthly, recurrent coding ambiguities were addressed by refining boundary rules. These included distinctions between adaptive deviation and unsafe drift, governance as practical decision support versus governance as documentation, intervention versus routine command decision and learning versus path-dependent reproduction of established routines. Inclusion and exclusion criteria were tightened iteratively to reduce category drift and improve consistency of application across meaning units.

Finally, coded meaning units were synthesised into a coupled heuristic model linking learning and institutional memory (E6), formal governance (E5), execution, deviation and drift (E1) and intervention (E4). This synthesis provided the basis for Figure 1 and for the interpretation of safety as a set of interdependent practices across the exercise lifecycle. The arrows in the model represent analytically proposed relationships between classes and should not be read as an empirically observed temporal sequence or as evidence of causal ordering.

Table 3 summarises interview-file support for each ethogram class. The table reports whether at least one extracted meaning unit from a readable interview file was coded to the class. It should therefore be interpreted as an indicator of evidentiary spread across the corpus, not as a measure of behavioural frequency, prevalence or incidence.

Saturation and evidentiary density

The analysis does not claim equal saturation across all four classes. Coverage was uneven: formal safety governance and risk work (E5) appeared across 36 respondents, deviation, drift and rule-bending (E1) across 13, learning, experience use and institutional memory (E6) across 12 and intervention shaping exercise trajectory (E4) across three.

This uneven distribution is treated as a finding about the interview corpus rather than as an estimate of prevalence or practical importance. E5 is the most strongly supported class. E1 and E6 are sufficiently recurrent to support analytical interpretation but remain vulnerable to reporting bias. E4 is retained because of its theoretical and practical importance, but should

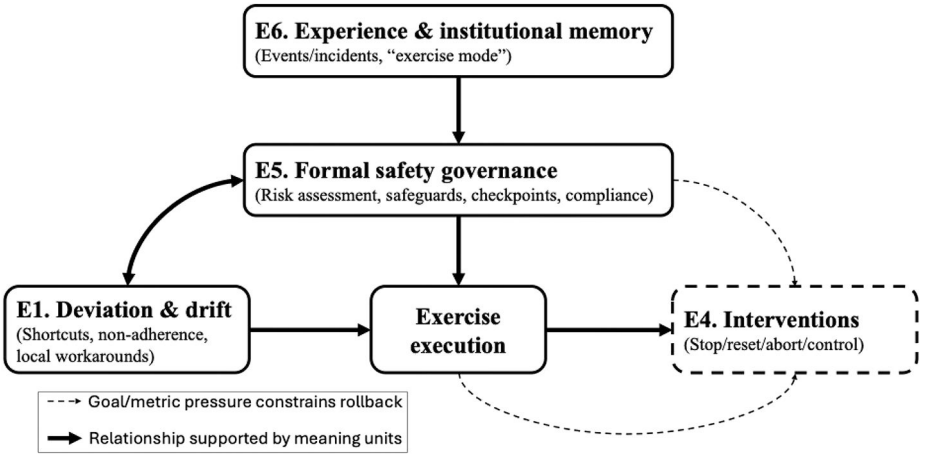


Figure 1. Proposed heuristic framework of interview-described safety-relevant practices in military exercises

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Table 3. Interview-file support by ethogram class

Ethogram class	Functional label	Interview files with evidence	*n*
E1	Deviation, drift and rule-bending	X04, X05, X07, X12, X14, X16, X20, X23, X30, X33, X35, X36, X40	13
E2	Unauthorised equipment use	None reported in extracted meaning units	0
E3	Unauthorised information use	None reported in extracted meaning units	0
E4	Intervention shaping exercise trajectory	X01, X36, X39	3
E5	Formal safety governance and risk work	X01, X02, X04, X05, X06, X07, X08, X09, X10, X11, X12, X13, X14, X15, X16, X17, X18, X19, X20, X21, X22, X23, X26, X27, X28, X29, X30, X31, X32, X33, X35, X36, X37, X38, X39, X40	36
E6	Learning, experience use and institutional memory	X02, X09, X14, X16, X22, X26, X27, X29, X30, X32, X36, X41	12

Note(s): Counts indicate the number of readable interview files containing at least one meaning unit coded to the class. They do not indicate behavioural frequency, prevalence or incidence. E2 and E3 were not reported in the extracted meaning units; this should not be interpreted as evidence that such behaviours were absent from exercise practice

be treated as empirically thinner and requiring prospective validation through observation, exercise-control records, safety logs and after-action materials.

Reporting bias and limits of interview-derived evidence

The interview-based design creates specific interpretive limits. In hierarchical military settings, accounts of safety may be shaped by professional identity, social desirability and concern about reputational, disciplinary or relational consequences. Formal safety governance may therefore be easier to discuss than rule-bending, unauthorised practice, intervention failure or learning breakdown. E5 may be overrepresented as institutionally legitimate safety talk, while E1, E4, unauthorised equipment use and unauthorised information use may be underreported.

This possibility was addressed by avoiding behavioural frequency claims, treating the ethogram as an interview-derived heuristic and reporting absent candidate classes as absent from the extracted meaning units rather than absent from exercise practice. Nevertheless, reporting bias remains a central limitation and motivates future triangulation.

Language and translation

The source interviews were conducted in Swedish. Excerpts reported in the manuscript were translated into English for publication. Translation prioritised preservation of meaning, pragmatic force and organisational sense rather than literal wording. Swedish originals were retained in the extraction trail to support translation transparency and checking of English renderings against the coded source material, consistent with methodological guidance on cross-language qualitative research (Squires, 2009).

Rigour and trustworthiness

Trustworthiness was supported through analytic traceability, proportional reporting, explicit coding rules and reflexive scrutiny. Reported claims were anchored to meaning-unit

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identifiers and interview file identifiers in the extraction table and only behaviour classes evidenced in the interview-derived meaning units were retained in the final framework. Operational definitions, inclusion and exclusion criteria and boundary rules were specified to reduce analytic slippage during directed coding. Negative and complicating cases were retained where they showed ambiguity between adaptive flexibility and unsafe drift or between governance as decision support and governance as documentary compliance.

Coding was conducted by the author alone and was not independently replicated by a second coder. This limits claims about coding reliability and places greater importance on transparency. To strengthen confirmability, interview-file support for each ethogram class is reported in Table 3 and the de-identified extraction table is provided as supplementary material, subject to confidentiality, OPSEC and deductive disclosure checks.

Researcher positioning and reflexivity

The researcher is used by the Swedish Armed Forces and holds a scientific position at the Swedish Defence University. Participants were not previously known to the researcher. Interviews were conducted in civilian clothing and participants were informed of the researcher's affiliation. This positioning supported access and contextual understanding, but required attention to power dynamics and their possible effects on recruitment, candour, interpretation and reporting.

Participant accounts were treated as situated accounts rather than direct measures of safety performance. Reflexive memoing documented analytical decisions, coding ambiguities and possible effects of organisational membership on interpretation, including whether familiarity with military terminology and routines made some assumptions appear self-evident.

Ethical considerations, confidentiality and data security

The study procedures were discussed with a university ethics advisor in 2018, who advised that formal ethical approval was not required because the project did not collect sensitive personal data and was assessed as minimal risk (Schüler, 2022b). Participation was voluntary and based on informed consent. In light of the hierarchical context, emphasis was placed on voluntariness, the possibility of declining without explanation and ensuring that participation status was not disclosed to supervisors or senior leadership.

Transcripts were de-identified during transcription and potentially identifying contextual details were removed or generalised. Reporting used file identifiers rather than personal identifiers and focused on behavioural classes and organisational practices rather than operationally sensitive specifics. Audio files and transcripts were stored on an external hard drive with access restricted to the author and kept in a locked safe when not in use. The data will be retained for 10 years in accordance with Swedish archiving legislation and applicable retention guidance (SFS 1990:782, 2026).

To reduce deductive disclosure risk in a bounded military organisation, analysis was intentionally delayed for three years. The supplementary extraction table should be reviewed before submission to ensure that quotations, paraphrases, role labels, local contexts and file identifiers do not create avoidable deductive disclosure or OPSEC risk.

Transferability and boundary conditions

The study is bounded by its Swedish military context, Swedish-language interview material and the researcher's organisational position. Military safety cultures, legal accountability frameworks, exercise-control arrangements and doctrines vary across national contexts,

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branches, coalitions and force structures. The Safety Ethogram should therefore be understood as analytically transferable rather than universally generalisable.

The framework may be useful in other military exercise settings because it identifies practice-level categories likely to recur in safety-critical training environments: governance, deviation, intervention and learning. However, the meaning and practical use of these categories require local adaptation and validation through observation, safety logs, exercise-control artefacts and after-action review materials.

Use of large language model

ChatGPT (GPT-5.4) was used for language editing, specifically grammar and spelling correction. It was not used to generate data, quotations, analyses or results. Responsibility for the final manuscript rests with the author.

Results

A safety ethogram for military exercises (E1, E4, E5, E6)

The analysis synthesised the interview material into 74 traceable meaning units (MU01-MU74), each linked to an interview file and coded using an ethogram logic for safety-relevant practices. Four interview-supported classes were identified: formal safety governance and risk work (E5), deviation, drift and rule-bending (E1), intervention shaping exercise trajectory (E4) and learning, experience use and institutional memory (E6). Together, these classes capture how participants described safety as planned, enacted, challenged, corrected and remembered across military exercise work.

Two candidate classes from the source ethogram, unauthorised equipment use and unauthorised information use, were not reported in the extracted meaning units and were therefore not included as empirically supported classes. This should not be interpreted as evidence that such behaviours are absent from military exercise practice. The retained classes should be read as interview-derived categories rather than direct observations of exercise behaviour or prevalence estimates. E5 was the most densely represented class, whereas E4 should be read as theoretically important but empirically thinner.

Figure 1 summarises the Safety Ethogram as a proposed coupled model in which E6 may inform E5, E5 may shape execution, E1 may destabilise the relationship between formal intent and practice and E4 may provide corrective stabilisation. The arrows indicate analytically proposed relationships, not an empirically observed temporal sequence or causal process.

E5. Formal safety governance as the dominant safety practice – risk work, documentation and structured planning

Across interviews, safety was repeatedly described as beginning in formal governance work: conducting risk assessments, identifying major risks and specifying protective measures (MU01). E5 includes routine planning practices such as continuous risk assessment and coordination with exercise leadership (MU14), embedding risk-management checkpoints in the planning cycle (MU15) and rehearsal or contingency planning with reserves and medical support (MU16).

Respondents also linked E5 to responsibility and traceability. Some emphasised that responsibility must be clearly assigned and supported by documentation (MU07), while others highlighted how rules and regulations were treated as a "regulatory part" of planning intended to prevent safety from "drawing the short straw" when tasks and resources were balanced (MU45). Safety governance was also described as institutionalised through

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routines intended to generate feedback loops, such as common templates routed to safety committees, with "feedback" and "learning" as explicit outputs (MU10).

At the same time, several meaning units showed important internal variation within E5. Formal systems could shift from decision-guiding governance to document-centred compliance. Risk analysis could be embedded in orders partly as "back-covering" rather than as a practice that shaped decisions (MU49) and safety routines could drift into "box-ticking" unless actively followed up (MU50). Similarly, some respondents described organisational compliance around risk analysis as becoming routine or mechanical because it had to be done (MU29), while traceability systems could exist mainly to collect documents rather than guide decisions (MU31).

E5 also extended beyond exercise conduct into organisational systems and acquisition. Participants described standardised safety requirements in procurement, such as "shall points" and "system safety plans" and the practical consequence that implementers may see only "a subset" rather than "the whole" (MU61). They also described ambiguity in how acceptance criteria were judged (MU62). Together, these accounts position E5 as a broad class spanning operational planning, administrative compliance, organisational governance and system-level safety requirements.

E1. Deviation and drift as a recurring pressure that decouples governance from practice

E1 captures accounts of behaviours and conditions through which actors depart from intended rules, plans and safety routines. Across the meaning-unit table, deviation was described both as explicit non-compliance, such as "many people do their own thing" (MU08) and as gradual drift, such as shortcuts that persist across contexts only to be problematised later: "you cut corners", "shortcuts", "quick fix" and "still a problem" (MU11).

Several meaning units located deviation in time pressure and uneven capacity. Participants contrasted "good" planning at staff level with field-level practice characterised as "go fast and wrong", noting that lower levels may not be given the time needed to comply (MU12). Others described sufficient time for exercise planning as rare despite the existence of safety regulations, resulting in a practical gap between formal requirements and what could be done (MU41). This decoupling was echoed in accounts of "lowering ambition", where dropping training elements could make plans "look good on paper" while undermining real competence and evaluation (MU42).

Deviation was also linked to mission and performance pressures. One meaning unit stated directly that "the drive to solve the task" clashes with safety regulations and that mission drive "takes over" (MU19). Another described how objectives had to be achieved, how rolling back "never happens" and how "officially we always have goal fulfilment" (MU21). In this logic, deviation was not necessarily framed as misconduct. It could also be described as a predictable outcome of goal regimes, tempo and organisational incentives that prioritise completion and the appearance of success.

The data also showed adaptive forms of deviation, where deviating was treated as skilled judgement rather than failure. Some respondents argued that personnel "must dare to break" or "put away the plan" when required (MU57), while others described deviation from rigid scheduling norms, such as not always ending at "16:30", as adaptive practice (MU74). A related form was local divergence from central tools through a homegrown "quick track" method: "we don't use" the standard tool, but "do a quick track" that had been developed locally (MU67).

Importantly, the data set included concrete illustrations of how rule-bending could create immediate safety risk or normalise unsafe assumptions. For example, an unsafe firing-range

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incident was described as turning "180 degrees the wrong way", with a loaded weapon and a fired shot, followed by the acknowledgement that "you know it's illegal" (MU04). Another meaning unit described how a safety matrix could exist but be undermined by an attitude that "it's an exercise so it's safe", with insufficient reflection by leaders in the moment (MU06). These accounts support treating E1 as a distinct class that affects the relationship between formal governance and practice even when E5 systems are present.

E4. Intervention as the corrective mechanism that reshapes execution when safety or learning is at risk

E4 was represented by fewer respondents than the other classes. It is therefore reported as an important but empirically thinner class rather than as a saturated category. Its inclusion is justified by its relevance to safety and learning validity, particularly where stop, reset, redirect or abort decisions shape the exercise trajectory.

E4 captures accounts of actions that stop, reset, redirect or abort exercise activity to preserve safety and/or learning validity. In the data set, the clearest marker was "red card" authority: stopping activity due to fatigue, expressed as "now you're going to sleep" and framing this as a consequence of poor planning (MU22). Red card was also explicitly described as a general stop-authority intervention tool (MU72).

Intervention was not described only as a stop mechanism, but also as a way to protect the exercise from collapsing into chaos or invalid learning conditions. Exercise control, described as "blue-and-yellow", was presented as crucial because "otherwise" there would be "chaos", enabling combat simulation while still following peacetime regulations (MU25). Relatedly, an exercise could be "broken" or aborted when imbalance made learning impossible (MU73).

E4 also appeared as continuous modulation during execution rather than only as discrete stop events. One respondent described moment-by-moment risk assessment and the need to "accelerate and brake" as psychosocial and physical risk fluctuated during activity (MU28). This positions intervention as a practical execution-phase safety practice that responds to dynamic risk conditions that cannot be fully controlled through pre-exercise planning.

Crucially, the data set also indicated constraints on intervention. As noted above, goal regimes could reduce the legitimacy of resetting or rolling back because rollback "never happens" and "officially" there is always "goal fulfilment" (MU21). Another meaning unit described pressure to proceed with "clever ideas" and the difficulty of "daring to say no" (MU23). Together, these meaning units suggest that E4 is not merely an available tool. It is shaped by organisational incentives and voice conditions that may determine whether intervention is enacted early, enacted late or avoided.

E6. Learning and institutional memory – when experience updates governance and when it becomes "old truths"

E6 captures accounts of how safety is shaped by experience, lessons and institutional memory. Several meaning units described learning as the basis for improving planning: using "old experience" and "new perspectives" as explicit inputs to safety planning (MU43) and reusing prior safety analyses for similar activities, as in "a shooting is a shooting", while adjusting when conditions differed (MU37). Respondents also described how safety cultures differed by branch, attributing mature aviation safety culture to historical accident experience and systematic learning and deviation work (MU46).

E6 was also described as institutionalised through formal learning systems. Participants described a safety-management-system orientation that built on capturing deviations and experience (MU52), a historical development of such systems from accidents, including

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reporting and trend analysis (MU53) and a learning loop with external investigators: "we get wiser" (MU56). Organisational memory was also represented through the growth of reporting lines and systems such as PRIO over time (MU64).

However, the data set indicated that learning and memory did not automatically improve safety; they could also create path dependence. Planning could be based on assumptions from "old truths" that were "not connected to what society looks like today" (MU17). Similarly, "exercise mode" was described as "putting on a kind of exercise jacket", where people learn to behave in a certain way during exercises because reality changes in that context (MU33). In other words, experience could train actors into exercise-specific behaviours that may or may not align with intended safety and realism.

E6 was also linked to drift. In a threat context, one respondent highlighted the risk of "routine drift", such as using the same routes and "skipping" important things (MU38). Another meaning unit described institutional "truth management" around risk framing, where claims that "soldiers will die unnecessarily" could later be treated as "nonsense" or covered over, pointing to how organisational narratives may shape what is learnable and what is officially acknowledged (MU27). These accounts support treating E6 as a class with both productive and problematic effects.

Integrating the ethogram: an evidence-based coupled model of safety in exercises

Taken together, the four interview-supported classes formed a proposed coupled model of safety work across the exercise lifecycle. The first linkage, E6 to E5, represents an ideal coupling in which experience, lessons and institutional memory may inform governance by revising risk assessments, planning assumptions and routine safety measures. The second linkage, E5 to execution, represents how governance may shape practice through planning, checkpoints, formal routines, leadership boundaries and resource arrangements. The third element, E1, represents recurrent pressures that may pull execution away from intended governance, including time scarcity, mission drive, local adaptation and goal regimes that normalise shortcuts or drift. The fourth element, E4, represents corrective stabilisation, that is, practices through which execution may be stopped, redirected, reset or aborted when safety and/or learning validity is threatened.

Discussion

Summary and theoretical positioning

The findings suggest that safety in military exercises is best understood as interdependent practice rather than as a single cultural attribute or administrative function. Across 74 traceable meaning units, four interview-supported classes were identified: formal safety governance and risk work (E5), deviation, drift and rule-bending (E1), intervention shaping exercise trajectory (E4) and learning, experience use and institutional memory (E6). Because the Safety Ethogram is derived from retrospective interview accounts, it should be read as a heuristic framework, not as direct observation, behavioural frequency measurement or evidence of causal sequence. E5 had the strongest support, E1 and E6 were sufficiently recurrent and E4 should be treated as theoretically important but empirically thinner.

The findings connect with safety-science debates on work-as-imagined and work-as-done, drift, adaptive capacity and resilience engineering (Dekker, 2011; Hollnagel, 2006; Hollnagel, 2014; Rasmussen, 1997). E5 captures safety as described through plans, risk assessments, documentation, responsibilities and organisational systems, whereas E1 captures how practice may move away from those arrangements through time pressure, mission drive, local adaptation, shortcuts or goal-fulfilment pressures. The framework does not replace existing safety-science concepts; rather, it translates them into a practice-oriented

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vocabulary for planning, debrief and after-action review, while retaining a workplace-learning focus on how practices become visible, discussable, legitimate and incorporated into future governance (Billett, 2001; Billett, 2004; Weick and Sutcliffe, 2007).

Governance, drift, intervention and learning

The dominance of E5 should be interpreted cautiously. Formal governance may be central to exercise safety, but it may also be the most institutionally legitimate way to discuss safety in interviews. Rule-bending, failed intervention, unauthorised practices and suppressed learning may be harder to narrate in a hierarchical setting. E5 is also internally diverse, including operational planning, administrative compliance and system-level governance. These forms may support safety, but they may also diverge when documentation demonstrates that safety has been considered without strongly shaping execution, consistent with organisational decoupling (Bromley and Powell, 2012; Meyer and Rowan, 1977).

Deviation should not be treated as a single moral or procedural category. Some departures from plan were described as shortcuts or erosion of safety margins; others were described as competent adaptation when the plan no longer fitted the situation. The central issue is whether departures were visible, justified, risk-aware, reviewable and open to learning. Adaptive flexibility becomes unsafe drift when departures are normalised without scrutiny, workarounds conceal deteriorating margins or official success narratives prevent examination of how task completion was achieved (Dekker, 2011; Vaughan, 1996).

Intervention is equally important, although less densely represented. E4 includes stopping, resetting, redirecting, aborting or modulating exercise activity when safety, realism or learning validity is threatened. Intervention is not only a last-resort safety measure; it is part of the learning infrastructure of an exercise. However, intervention is structurally uneven: senior controllers, commanders, safety officers and junior participants occupy different positions in relation to stop or reset authority. Goal-fulfilment pressure, professional identity, peer expectations, trust and fear of negative judgement may discourage early intervention. Intervention capacity therefore cannot be reduced to formal stop authority; it must be legitimised, rehearsed and protected as part of exercise design (Cannon and Edmondson, 2005; Carroll *et al.*, 2002).

Learning and institutional memory also have a dual role. In its productive form, E6 updates governance by translating experience into revised risk assessments, planning routines, stop criteria or control arrangements. However, institutional memory may also preserve outdated assumptions, "old truths" or exercise-specific habits that no longer fit current conditions. Reporting systems and after-action processes may identify lessons without ensuring implementation, leaving the practical updating loop from E6 to E5 weak (Argyris, 1976; Klammer and Gueldenberg, 2019; Maccioni *et al.*, 2024).

Mechanisms sustaining the governance-practice gap

Taken together, the findings suggest that the governance-practice gap is sustained by several mechanisms: tempo and mission primacy make task completion a dominant performance signal; hierarchy shapes what can be voiced; documentary accountability may substitute for practical decision support; official success narratives may discourage rollback or acknowledgement of failure; and trust, reputation, peer pressure and fear of negative judgement may limit disclosure and intervention. These mechanisms help explain why organisations may formally value safety while also rewarding continuity, completion and the appearance of successful execution (Brunsson, 2019; Meyer and Rowan, 1977; Bromley and Powell, 2012).

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Implications for workplace learning and practice

For workplace-learning research, the framework shows how safety learning depends on whether safety-relevant practices are made visible, deviations are examined rather than merely judged, intervention is legitimate and usable and experience is translated into future governance (Billett, 2001; Billett, 2004; Dewey, 1998). It also shows that learning may be distorted: personnel may learn that shortcuts are tolerated, rollback is unlikely, official success matters more than reflective accuracy or some safety concerns are safer to leave unspoken.

For practice, the Safety Ethogram should be used as a structured debrief and observation aid rather than as a scoring instrument. Before an exercise, E5 can be used to examine whether governance has been translated into responsibilities, decision points, stop criteria, escalation routes and feedback mechanisms. During execution, E1 can guide attention to where practice diverges from formal expectations, while E4 can support examination of whether intervention was available, considered, delayed, enacted or avoided. After the exercise, E6 can help assess whether experience is translated into revised assumptions, planning routines or governance mechanisms.

Limitations and future research

Several limitations follow from the design. The study is based on retrospective interview accounts rather than direct observation. Formal governance may therefore be overrepresented, while deviation, intervention, unauthorised equipment use, unauthorised information use and learning breakdown may be underreported because of reputational, disciplinary or relational risk. The four classes also have unequal evidentiary density, with E4 especially requiring further validation. Coding was conducted by an organisational insider and would be strengthened in future work by independent coding or external audit (Elo and Kyngas, 2008; Braun and Clarke, 2019). Finally, the study is bounded by its Swedish military context and should be treated as analytically transferable rather than universally generalisable.

Conclusion

Military exercises are safety-critical workplace-learning environments in which personnel plan, adapt, intervene and learn under conditions of tempo, hierarchy and evaluation pressure. This study developed an interview-derived Safety Ethogram identifying four interlinked classes of practice: formal safety governance and risk work (E5), deviation, drift and rule-bending (E1), intervention shaping exercise trajectory (E4) and learning, experience use and institutional memory (E6).

The study contributes a bounded framework for examining how safety is described, challenged, corrected and remembered in military exercise work. Its value lies not in direct behavioural observation, prevalence estimates or a universal causal model, but in offering a structured vocabulary linking workplace learning and safety science. For practitioners, the framework can support more disciplined observation, debrief and after-action review by focusing attention on how safety was governed, where drift emerged, whether intervention was legitimate and usable and what experience should change future planning.

Because the framework is based on retrospective interview accounts from a Swedish military context, it should be treated as analytically transferable rather than universally generalisable. Future research should validate it through observation and triangulation with safety logs, exercise-control artefacts, and after-action review material.

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Supplementary material

The supplementary material for this article can be found online.

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